

Task 1 — Selection and Research

Designing a Jurisdiction-Aware AI Credit Scoring Governance Tool for a Cross-Border Bank

1. Introduction

The increasing adoption of artificial intelligence (AI) and machine learning in retail banking has transformed the way financial institutions assess creditworthiness and process lending decisions. Cross-border banks are increasingly using AI-driven credit scoring systems to improve operational efficiency, reduce costs, accelerate loan approvals, and enhance risk management. However, the use of AI in consumer lending has also created significant regulatory and governance challenges, particularly when financial institutions operate across multiple jurisdictions with differing regulatory priorities.

This project proposes the design of a jurisdiction-aware AI credit scoring governance tool for a cross-border bank. The tool is designed to monitor fairness risks, explainability requirements, model governance obligations, and regulatory compliance outcomes under different legal regimes. Rather than functioning as a loan approval model itself, the proposed solution acts as a governance and compliance monitoring layer surrounding an existing AI credit scoring model.

The project focuses on the regulatory divergence between the United States and the European Union. Both jurisdictions are increasingly concerned about the use of AI in financial services, but they approach the issue differently. The United States primarily emphasizes fair lending, anti-discrimination obligations, and adverse action explanations, while the European Union places stronger emphasis on high-risk AI governance, human oversight, transparency, and lifecycle risk management under the EU AI Act.

The proposed tool demonstrates how a RegTech solution can translate differing regulatory requirements into configurable monitoring rules, governance workflows, and compliance outputs.

2. Selected Entity: HSBC

This project uses HSBC Holdings plc as the selected regulated entity.

HSBC is one of the largest international banking groups in the world, operating across Asia, Europe, North America, and the Middle East. The bank provides retail banking, wealth management, commercial banking, and global markets services across multiple jurisdictions, including both the United States and the European Union.

HSBC represents a suitable case study for this assignment for several reasons.

First, HSBC operates across multiple major regulatory environments, making it highly exposed to cross-jurisdictional regulatory divergence. This creates a realistic need for configurable compliance and governance systems.

Second, large international banks are increasingly adopting AI and machine learning technologies in areas such as credit scoring, fraud detection, anti-money laundering monitoring, and customer onboarding. As a result, these institutions face growing pressure to demonstrate fairness, explainability, governance, and model accountability.

Third, global banks face substantial operational complexity when attempting to harmonize AI governance standards across jurisdictions. Regulatory expectations in one jurisdiction may not fully align with those in another, requiring governance systems that can dynamically adapt to changing legal requirements.

The proposed tool is therefore designed as a governance layer that could realistically support a global banking institution such as HSBC.

3. Domain Selection: AI Credit Scoring Governance

The selected domain for this project is AI credit scoring governance.

AI-based credit scoring systems use machine learning models to evaluate the creditworthiness of loan applicants based on financial, demographic, and behavioral information. Compared to traditional rule-based scoring systems, AI models can process larger datasets and identify complex statistical relationships that may improve predictive performance. However, the use of AI in credit decisions also introduces significant governance and compliance risks.

These risks include:

- Discriminatory lending outcomes;
- Bias against protected groups;
- Lack of explainability;
- Inability to justify adverse decisions;
- Model drift over time;
- Data quality risks;
- Governance failures;
- Excessive automation without human oversight.

Because lending decisions directly affect consumers' financial access and economic

opportunities, regulators increasingly view AI credit scoring as a high-risk use case requiring enhanced governance and monitoring.

The proposed tool therefore focuses on monitoring:

- Fairness metrics;
- Approval rate disparities;
- False positive and false negative differences;
- Explainability requirements;
- Model drift;
- Governance escalation triggers;
- Human oversight obligations.

4. United States Regulatory Approach

4.1 Fair Lending and Consumer Protection

The United States regulatory approach toward AI credit scoring primarily focuses on fair lending, anti-discrimination obligations, consumer protection, and adverse action explanation requirements under existing consumer finance laws and regulatory guidance.

US regulators are particularly concerned about the possibility that AI models may produce discriminatory outcomes against protected groups, even when models do not explicitly use protected attributes such as race or gender. A key regulatory concern is disparate impact, where seemingly neutral models produce significantly different outcomes across demographic groups. The Consumer Financial Protection Bureau (CFPB) has emphasized that lenders using AI-based systems remain subject to existing fair lending laws and adverse action notice requirements under the Equal Credit Opportunity Act (ECOA) and Regulation B. CFPB guidance states that creditors using complex algorithms are still required to provide specific and accurate reasons for adverse credit decisions rather than relying on the opacity of AI systems.

This creates several governance implications for financial institutions:

- AI models must remain explainable;
- Rejected applicants must receive understandable explanations;
- Fairness outcomes must be monitored continuously;
- Potential discriminatory effects must be investigated.

The United States approach therefore focuses heavily on fairness outcomes and consumer protection.

4.2 Key Regulatory Themes in the US

The main regulatory priorities reflected in the proposed tool include:

- Fair lending monitoring;
- Adverse action explanation requirements;
- Disparate impact analysis;
- Consumer protection;
- Ongoing fairness review.

Within the proposed governance tool, these requirements are translated into:

- Approval disparity monitoring;
- Rejection reason generation;
- Fair lending alerts;
- Escalation workflows;
- Compliance reporting outputs.

5. European Union Regulatory Approach

5.1 High-Risk AI Governance

The European Union approaches AI regulation through a broader governance and lifecycle risk management framework. Under the EU AI Act, AI systems used to evaluate the creditworthiness of natural persons or establish credit scores are generally treated as high-risk AI systems, subject to limited exceptions, because they can significantly affect individuals' access to essential financial services. This classification creates enhanced governance obligations for organizations deploying such systems.

Compared to the United States, the EU places stronger emphasis on:

- Human oversight;
- Transparency;
- Governance documentation;
- Risk management systems;
- Data governance;

- Continuous monitoring;
- Lifecycle accountability.

The EU regulatory framework is therefore more governance-oriented and process-oriented rather than focusing exclusively on lending outcomes.

5.2 Key Regulatory Themes in the EU

The main EU regulatory priorities reflected in the proposed tool include:

- High-risk AI governance;
- Mandatory human oversight;
- Transparency obligations;
- Bias testing;
- Documentation requirements;
- Post-deployment monitoring.

Within the proposed governance tool, these obligations are translated into:

- Enhanced monitoring workflows;
- Human review escalation;
- Governance documentation generation;
- Bias monitoring dashboards;
- Drift monitoring systems;
- Risk classification outputs.

6. Regulatory Divergence Between the US and EU

Although both the United States and the European Union are concerned about the risks of AI in credit scoring, the two jurisdictions prioritize different governance objectives. The United States primarily focuses on preventing discriminatory lending outcomes and ensuring that consumers receive understandable adverse action explanations under fair lending and consumer protection frameworks. In contrast, the European Union adopts a broader governance framework that emphasizes lifecycle accountability, human oversight, transparency, and formal risk management processes. As a result, the same AI model may trigger different governance requirements depending on the jurisdiction.

For example, a fairness metric breach in the United States may trigger a fair lending

investigation and review of adverse action explanations. Under the EU AI Act, the same fairness issue may additionally require:

- Human oversight escalation;
- Enhanced governance documentation;
- Risk mitigation procedures;
- Post-market monitoring actions.

This regulatory divergence creates operational challenges for global financial institutions and demonstrates the need for jurisdiction-aware RegTech systems.

7. Why a Jurisdiction-Aware RegTech Tool Is Necessary

Traditional compliance approaches often rely on spreadsheets, static documentation, or fragmented manual review processes. These approaches are increasingly insufficient for managing AI governance risks across multiple jurisdictions. Several challenges make a jurisdiction-aware governance tool necessary. First, regulatory requirements evolve frequently and differ significantly across jurisdictions. Second, AI models can drift over time, causing fairness and performance metrics to deteriorate after deployment. Third, large international banks operate multiple AI systems simultaneously across business lines and geographic regions. Fourth, governance expectations increasingly require continuous monitoring rather than periodic manual review.

The proposed tool addresses these challenges by translating regulatory requirements into configurable monitoring rules and governance workflows.

The tool is therefore intended to support:

- Cross-jurisdictional governance consistency;
- Fairness monitoring;
- AI risk management;
- Compliance escalation;
- Governance documentation;
- Human oversight processes.

8. References

United States

Consumer Financial Protection Bureau (CFPB) <https://www.consumerfinance.gov/about->

[us/newsroom/cfpb-issues-guidance-on-credit-denials-by-lenders-using-artificial-intelligence/](#)

Consumer Financial Services Law Monitor – CFPB Regulation B Discussion

[https://www.consumerfinancialserviceslawmonitor.com/2026/04/cfpb-finalizes-regulation-b-subpart-a-rule-largely-as-proposed/](#)

European Union

European Parliament – EU AI Act Overview [https://www.europarl.europa.eu/Reg-DATA/etudes/ATAG/2025/769509/EPRS_ATA\(2025\)769509_EN.pdf](#)

Selected Entity

HSBC Official Website [https://www.hsbc.com/](#)

HSBC Holdings plc Annual Report and Accounts 2025 [https://www.hsbc.com/investors/results-and-announcements/annual-report](#)

HSBC in the USA official site [https://www.about.us.hsbc.com/](#)

EUR-Lex official text of Regulation (EU) 2024/1689 [https://eur-lex.europa.eu/eli/reg/2024/1689/oj/eng](#)